VLM benchmark: Qwen2.5-VL 3B vs 7B

MRIModelBench · full test set, 1,584 images · notebooks/ollama_vlm.ipynb · 7B run on Colab

In short

- The 7B model is more accurate than the 3B model: **0.491 vs 0.426** zero-shot and **0.480 vs 0.396** few-shot (about +6 and +8 points; the margin of error is ±2.5 points).
- **Macro F1 did not improve**: 0.357 vs 0.371 zero-shot, 0.369 vs 0.341 few-shot. These differences are within noise. Both models stay far below the Decision Tree (0.77) and above random guessing (0.25).
- The bias moved, it did not go away. The 3B model over-answered **glioma** (57% of zero-shot answers). The 7B model over-answers **pituitary** (54%) and **notumor**. Meningioma is still almost never found (recall 0.00 zero-shot, 0.06 few-shot).
- Tumours called healthy: 23% (zero-shot) and 32% (few-shot) for 7B, against 12% and 54% for 3B and 10% for the Decision Tree.

1. What the two runs were

	qwen2.5vl:3b	qwen2.5vl:7b
Hardware	Laptop, 4 GB GPU, partly on CPU	Colab T4, 15 GB GPU
Run time (zero-shot / few-shot)	82 / 116 min	70 / 68 min (4191 s / 4082 s)
Images, setups	Same 1,584 test images; zero-shot and few-shot (2 examples per class)	
Settings	Same prompt, temperature 0, seed 0, JSON-schema-constrained answer, context 12,288	
Failed calls	0	0 / 1,584 in both setups

Run times are not comparable, because the hardware is different. Only the answers are compared here.

2. Headline results

Setup	Model	Accuracy	±95%	Macro F1	Tumours called healthy
Zero-shot	3B	0.426	0.024	0.371	146 / 1,184 (12.3%)
	7B	**0.491**	0.025	0.357	267 / 1,184 (22.6%)
Few-shot	3B	0.396	0.024	0.341	641 / 1,184 (54.1%)
	7B	**0.480**	0.025	0.369	379 / 1,184 (32.0%)
Random guess / Decision Tree		0.25 / n/a		0.25 / 0.77	Tree: 122 (10.3%)

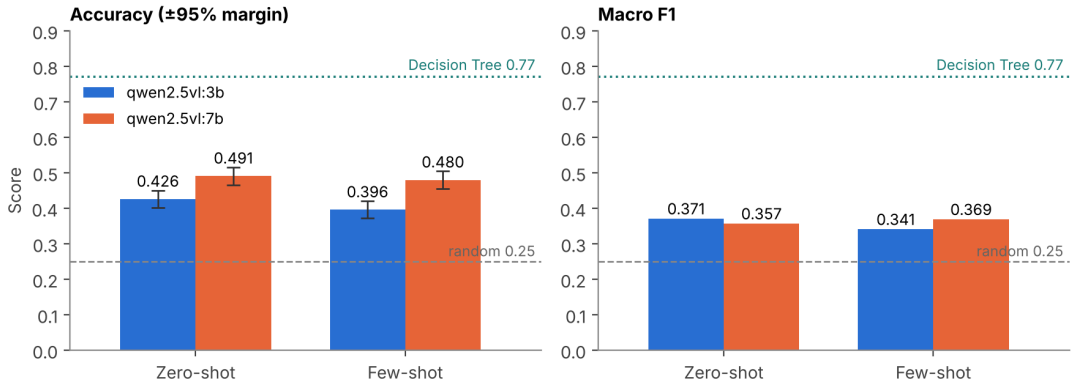


Figure 1. Accuracy has error bars; macro F1 does not, but differences of 0.01 to 0.03 should be read as noise.

Why accuracy rose but macro F1 did not. Macro F1 averages the four classes equally. The 7B model got more right overall, mostly by finding healthy scans and pituitary tumours. It still misses glioma and meningioma, and those two classes pull the average down.

3. Did the patterns hold? Per class

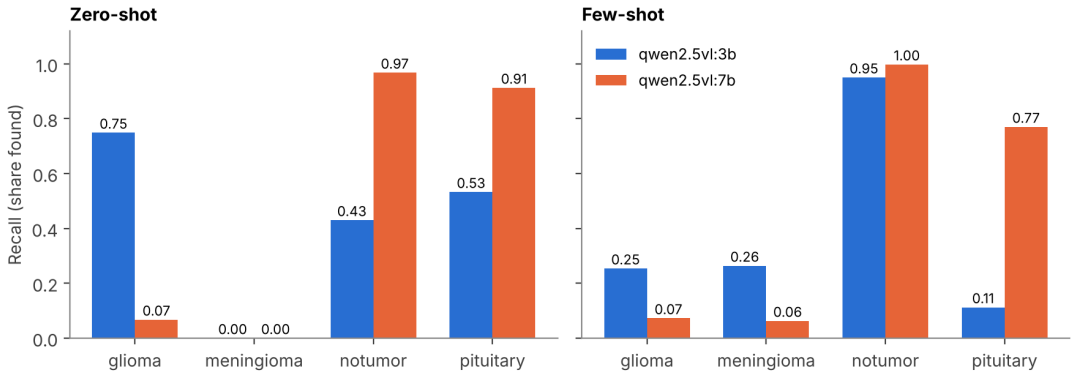


Figure 2. Recall: the share of each true class the model found.

F1 per class	3B zero	7B zero	3B few	7B few
glioma	0.449	0.113	0.310	0.122
meningioma	0.000	0.000	0.329	0.116
notumor	0.479	0.734	0.535	0.677
pituitary	0.557	0.581	0.189	0.561

- **Glioma collapsed.** The 3B model found 75% of gliomas zero-shot; the 7B model finds 7%.
- **Notumor and pituitary improved** (notumor recall 0.97 and 1.00; pituitary 0.91 and 0.77), and this is where the accuracy gain comes from.
- **Meningioma stayed weak.** Zero-shot it is never predicted by either model. Few-shot, the 7B model is precise when it does say meningioma (0.74), but it says it only 2% of the time.

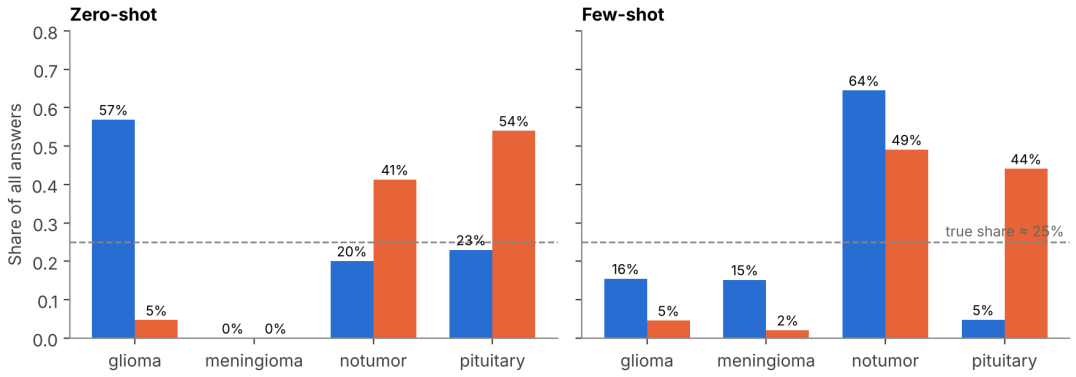


Figure 3. Share of all answers per class. A model with no class bias would sit near the dashed 25% line.

Share of answers	3B zero	7B zero	3B few	7B few
glioma	56.9%	4.7%	15.5%	4.6%
meningioma	0.0%	0.0%	15.2%	2.1%
notumor	20.1%	41.3%	64.5%	49.1%
pituitary	23.0%	54.0%	4.8%	44.1%

4. Tumours called healthy

This is the error that matters most for a screening-style task: a scan with a tumour that the model calls `notumor`.

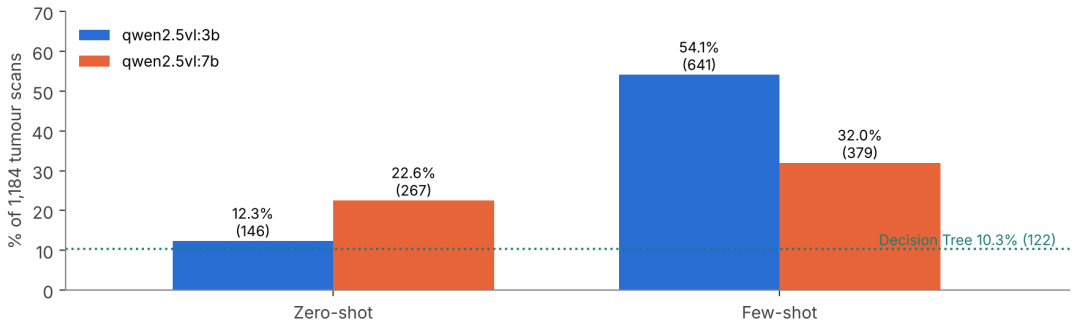


Figure 4. Share of the 1,184 tumour scans predicted as notumor.

The 7B model is worse than 3B zero-shot (23% vs 12%) but much better than 3B few-shot (32% vs 54%). Neither VLM matches the Decision Tree, which misses 10.3%.

5. Do the two models agree?

Setup	Both correct	Only 7B correct	Only 3B correct	Both wrong
Zero-shot	395	383	279	527
Few-shot	448	312	180	644

The models often disagree. The 3B model is right where 7B is wrong on 279 images (zero-shot), so the larger model is not a superset of the smaller one. It simply has a different bias.

6. Possible explanations and caveats

- **General-purpose model.** Qwen2.5-VL was not trained for MRI classification. A larger model has more general knowledge, but not more tumour-specific knowledge. This is a hypothesis; we did not test it.
- **Forced choice.** The schema forces one of four labels, so any uncertainty comes out as a bias toward whichever label the model prefers.
- **Different hardware.** The runs used different GPUs. With temperature 0 and a fixed seed we expect similar behaviour, but small numeric differences are possible.
- **Single seed, single prompt.** Run-to-run noise was about 1 point in the 3B pilot study, and we did not repeat the 7B run.
- **No 7B pilot.** Unlike the 3B model, the 7B model was run on the full set only, so we have no pilot-versus-full comparison for it.
- **Notebook text is stale.** The Conclusions cell in `ollama_vlm.ipynb` still describes the 3B run only; its 7B table is correct.

7. What this means for the project report

- Quote the full-test-set numbers for both models, not the 200-image pilot.
- Say that the larger VLM improved accuracy by about 6 to 8 points, which is larger than the margin of error, but did not improve macro F1, where the differences are within noise.
- Describe the change in bias (glioma to pituitary and notumor) rather than saying 7B is simply better.
- The Decision Tree is ahead by about 0.40 macro F1 over either VLM. The VLM comparison supports the point that a small trained model beats a general-purpose zero-shot model on this task.
- Fix the meningioma/glioma confusion wording in the notebook (the 87 errors are glioma predicted as meningioma).